



MURUGAPPA POLYTECHNIC COLLEGE

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(An Academically Autonomous Institution)

DEPARTMENT OF COMPUTER ENGINEERING

REAL TIME CRASH DETECTION & INSTANT EMERGENCY ALERT WITH LIVE LOCATION TRACKING

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PRESENTATION OUTLINES

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ABSTRACT

Road accidents in rural and remote areas often go unnoticed, causing delays in emergency response and increasing fatality rates. To address this issue, we developed a Real Time Crash Detection and Instant Emergency Alert System using Artificial Intelligence and computer vision. The system uses a webcam and YOLOv8 model to detect vehicle collisions in real time by analyzing vehicle movement and bounding box overlap. When an accident is detected, it automatically sends SMS and email alerts with the exact latitude, longitude, and a Google Maps navigation link. The system also logs accident details into a monitoring dashboard. This cost-effective solution eliminates the need for expensive vehicle sensors and aims to reduce response time, improve rescue efficiency, and enhance road safety.

LITERATURE SURVEY

- Sensor-based accident detection systems
- GPS & GSM-based vehicle alert systems
- Smart vehicle embedded modules
- AI-based traffic monitoring research

OBJECTIVES

- Develop a real-time AI-based crash detection system
- Implement YOLOv8 for vehicle detection
- Detect collisions using motion and overlap analysis
- Generate instant SMS and email alerts
- Share Google Maps live location
- Maintain accident logging dashboard
- Reduce emergency response time
- Provide low-cost scalable solution

NOVELTY OF THE SYSTEM

- Camera-based accident detection without vehicle sensors
- Real-time AI processing using YOLOv8
- Collision detection using bounding box overlap and motion tracking
- Automatic multi-channel alert system (SMS + Email)
- Google Maps navigation link for instant location access
- Integrated accident monitoring dashboard
- Low-cost implementation suitable for rural areas

EXISTING SYSTEMS

- Currently, accident detection is done using:
- In-vehicle sensor-based systems
- GPS & GSM modules inside vehicles
- Manual accident reporting
- Smart vehicle embedded systems

Limitations:

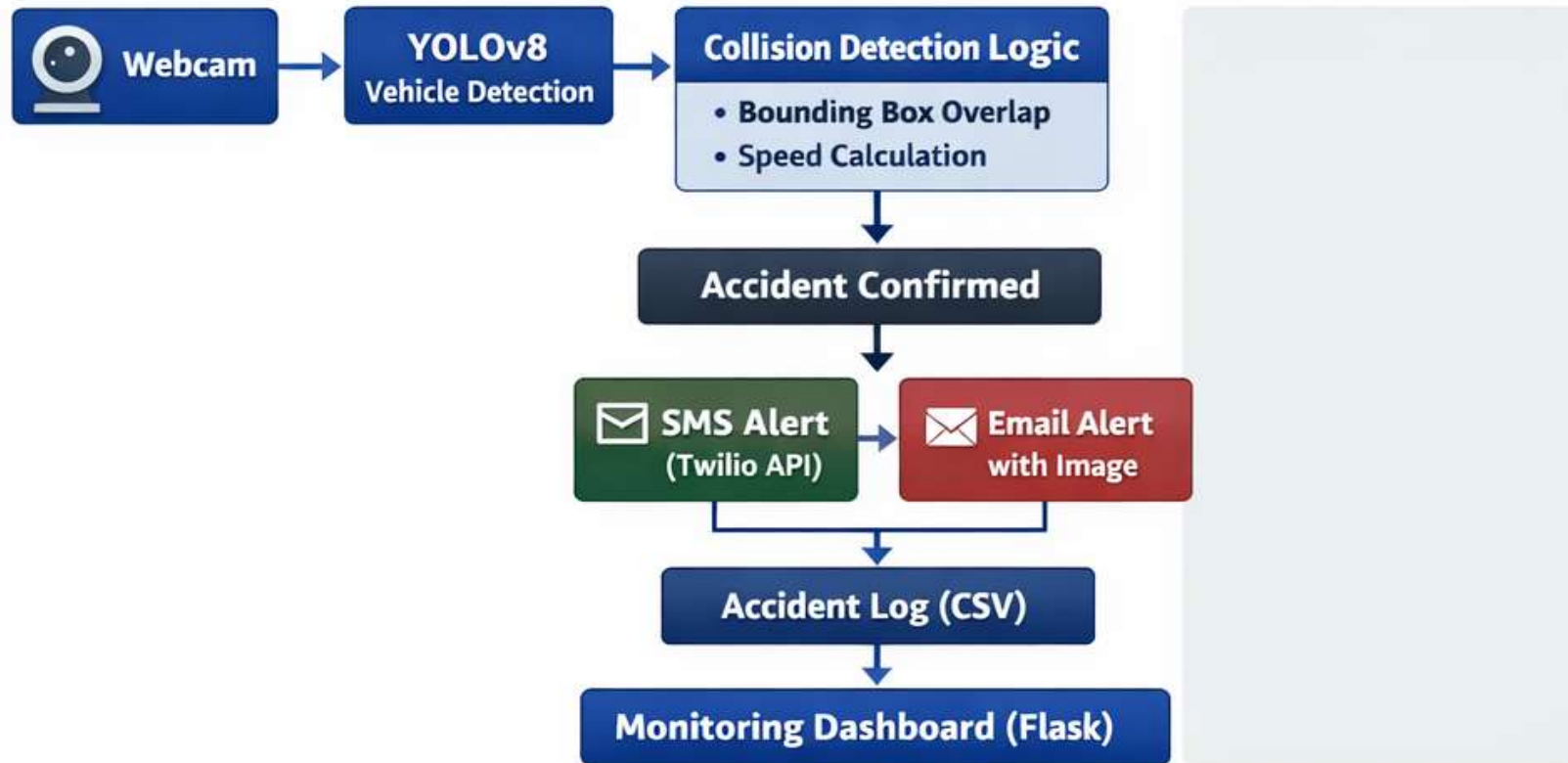
- Expensive hardware
- Only available in smart vehicles
- Not scalable in rural areas
- Delay in reporting if no witnesses

PROPOSED SYSTEMS

- Webcam-based monitoring
- YOLOv8 vehicle detection
- Collision detection using overlap logic
- Instant SMS & Email alerts
- Dashboard monitoring

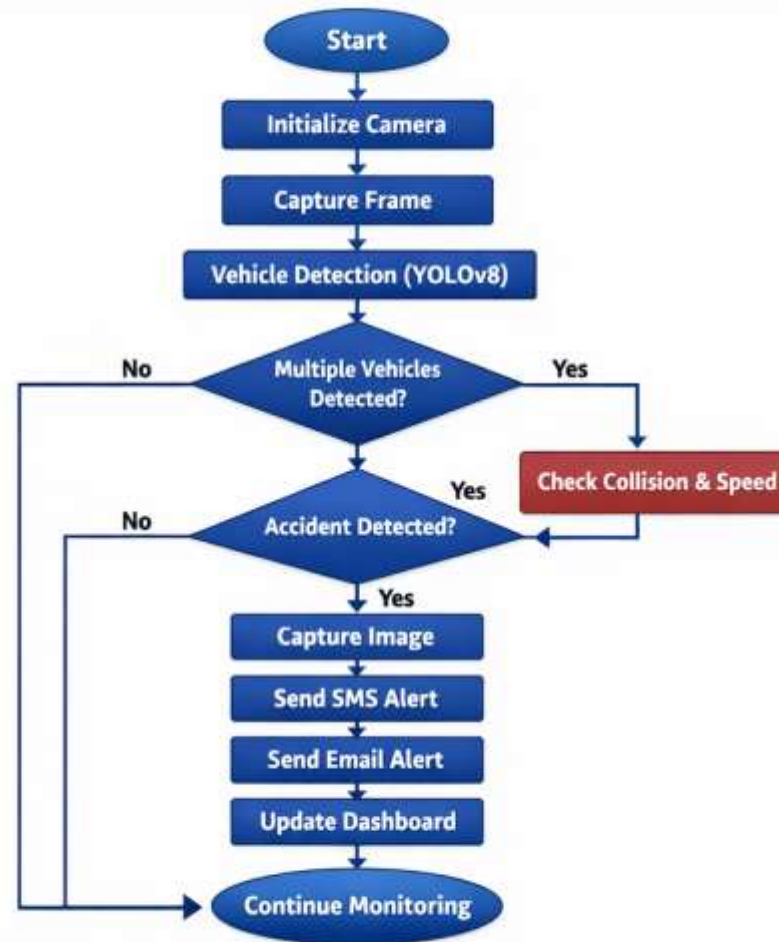
BLOCK DIAGRAM

System Block Diagram



FLOWCHART

System Flowchart



SOFTWARE DETAILS

Software:

- Python 3.11
- OpenCV
- YOLOv8
- Flask
- Twilio API
- Gmail SMTP

Hardware:

- AMD RYZEN 3
- 8 GB RAM
- Webcam

RESULTS AND DISCUSSION

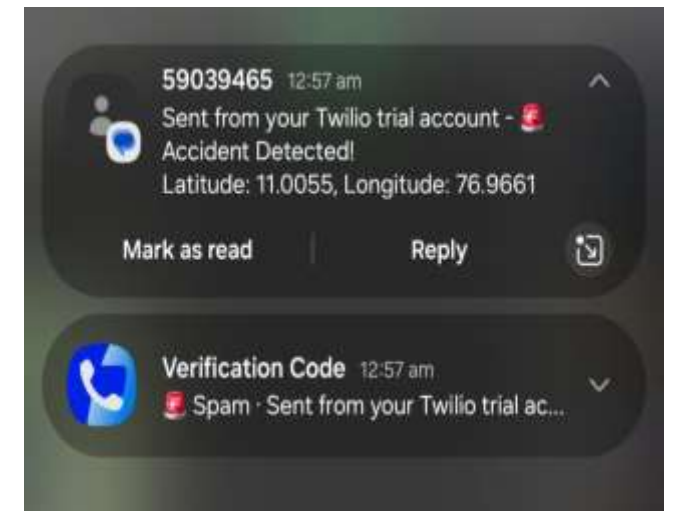
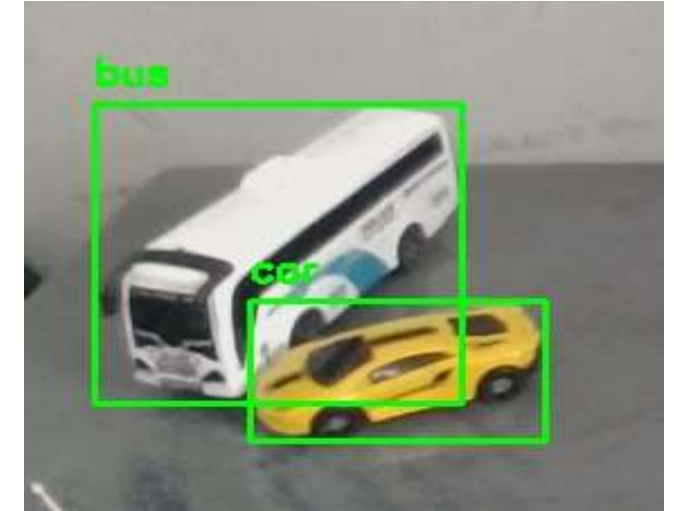
Total Accidents

6

Today's Accidents

4

Time	Latitude	Longitude	Google Map	Image
2026-02-16 23:06:18	Demo_Latitude	Demo_Longitude	Open Map	
2026-02-16 23:26:33	Demo_Latitude	Demo_Longitude	Open Map	
2026-02-17 00:33:03	13.1234	80.1234	Open Map	
2026-02-17 00:33:31	13.1234	80.1234	Open Map	
2026-02-17 00:33:59	13.1234	80.1234	Open Map	
2026-02-17 00:45:01	13.1234	80.1234	Open Map	



RESULTS AND DISCUSSION



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\ELCOT\OneDrive\Desktop\Crash_Project> python main.py
raise ConnectionError(err, request=request)
requests.exceptions.ConnectionError: ('Connection aborted.', RemoteDisconnected('Remote end closed connection without re
sponse'))
PS C:\Users\ELCOT\OneDrive\Desktop\Crash_Project> python main.py
System Started... Waiting for Collision
IMPACT DETECTED EXACT FRAME
SMS Sent to all numbers
Email Sent with Image
Accident logged to CSV
IMPACT DETECTED EXACT FRAME
SMS Sent to all numbers
Traceback (most recent call last):
Ln 30, Col 24  Spaces: 4  UTF-8
```

TIME CONSUMPTION

Performance Analysis:

- Real-time vehicle detection using YOLOv8
- **Frame processing:** < 1 second
- **Collision confirmation:** Instant after overlap detection
- **SMS alert generation:** Within 3–5 seconds
- **Email alert with image:** Within few seconds
- **Dashboard update:** Immediate

ADVANTAGES

- Fully automated accident detection
- No dependency on vehicle sensors
- Low-cost implementation (~₹6000)
- Real-time monitoring and alert system
- Multi-channel communication (SMS + Email)
- Centralized dashboard logging
- Scalable for highways and smart cities

APPLICATIONS

- National & State Highways
- Toll Plazas
- Smart City Surveillance
- Rural & Remote Road Monitoring
- Traffic Control Rooms
- Police & Emergency Response Centers

CONCLUSION AND FUTURE ENHANCEMENT

Conclusion:

- The proposed system successfully demonstrates a real-time AI-based accident detection and emergency alert mechanism using computer vision. By integrating YOLOv8, motion analysis, and instant communication services, the system automatically detects collisions and sends alerts with exact location details. This solution is cost-effective, scalable, and capable of reducing emergency response time, thereby improving road safety.

Future Enhancement:

- Integration with cloud-based database
- Mobile application for authorities
- Deployment on highway CCTV infrastructure
- Multi-camera angle monitoring system
- Automatic recording of 5 seconds before and after impact
- Sending accident video clip along with image in email and mobile alerts

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